FYS-STK3155/4155 October 19

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Poisson's eq in 1-dim

$x \in [0,1]$

$$-u''(x) = -\frac{d^2u}{dx^2} = f(x)$$

in examples:

$$f(x) = (3x + x^2)\exp(x)$$

$$u(x) = x(1-x)\exp(x)$$

classical solution (Thomas algorithm)

Discretize $x_i = x_0 + i \cdot \Delta x$

$\vec{x}^T = [x_0, x_1, \ldots x_n]$

$i = 0, 1, 2, \ldots n$ ($n+1$ points)

$$\Delta x = \frac{x_n - x_0}{n}$$

$u(x) \Rightarrow u(x_i) = u_i$

$f(x) \Rightarrow f(x_i) = f_i$

$$\frac{d^2u}{dx^2} = \frac{u(x_i+\Delta x)+u(x_i-\Delta x)-2u(x_i)}{\Delta x^2}$$

$$+ \mathcal{O}(\Delta x^2)$$

Poisson's eq :

$$-\left(u_{i+1} + u_{i-1} - 2u_i\right) = \Delta x^2 f_i$$

$$\tilde{f}_i = f_i$$

Boundary conditions

(BC) $\quad u(0) = u_0 = u(1) = u_n = 0$

$n = 1$

$$-(u_2 + u_0 - 2u_1) = \tilde{f}_1$$

$$-(u_3 + u_1 - 2u_2) = \tilde{f}_2$$

$$\vdots$$

$$-(u_m + u_{m-2} - 2u_{m-1}) = \tilde{f}_{m-1}$$

$$\vec{u}^T = [u_1, u_2, \dots u_{m-1}]$$

$$A = \begin{bmatrix} +2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & -1 & 2 & -1 & \\ & & -1 & \ddots & \\ & & & & 2 \end{bmatrix} \Rightarrow A\vec{u} = \vec{f}$$

with a neural network (NN)

– Hard approach

Trial $u_t(x;\theta) = ?$

$\theta = \{W, b\}$

$NN = N_\theta(x)$

$$u_t(x;\theta) = \underbrace{x(1-x)}_{BC} N_\theta(x)$$

$$C_\theta(x) = \frac{1}{n}\sum_i \left(-u_t''(x_i) - f_i\right)^2$$

$$\vec{\theta} = \underset{\theta}{\operatorname{argmin}} \; C_\theta(x)$$

Soft

$$C_\theta(x) = \frac{1}{n} \sum_i \left(-u_\theta''(x_i) - f_i \right)^2$$

Add Boundary

$$N_\theta(x) \quad x = 0 \;\wedge\; x = 1$$

$$C_{BC}(x) = \frac{\lambda_{BC}}{2} \left(N_\theta^2(0) + N_\theta^2(1) \right)$$

$$e_0 = N_e(x=0) - u(x=0)$$

$$= 0$$

$$= N_e(x=0) = N_\theta(0)$$

$$e_1 = N_\theta(1) - u(1) = N_\theta(1)$$

$$\frac{1}{2}\left(e_0^2 + e_1^2\right)$$

$$u_0 = a \quad 1 \quad u_1 = b$$

$$e_0 = N_\theta(0) - a$$

$$e_1 = N_\theta(1) - b$$

$$\hat{\epsilon} = \underset{\epsilon}{\arg\min} \left(C_\epsilon(x) + \frac{\lambda_{BC}}{2} \times \left(\nu_\epsilon^2(0) + \nu_\epsilon^2(1) \right) \right)$$

$$C_\epsilon(x) = \frac{1}{n} \sum_i \left(-\nu_\epsilon''(x_i) - f_i \right)^2$$

λ_{BC} is a hyperparameter.

Confusion matrix (Binary case)

	Predicted positive	Predicted negative
positive	TP	FN
negative	FP	TN

TP = True positive { sick correct class.

$$acc = accuracy =$$

$$\frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{1}{n} \sum_i I(y_i = \tilde{y}_i)$$

TPR = True positive rate
(fraction of actual positives
correctly classified)
Sensitivity

$$TPR = \frac{TP}{TP + FN}$$

FPR = False alarm rate the fraction of the actual negatives incorrectly classified as positives

$$FPR = \frac{FP}{FP + TN}$$

True negative rate

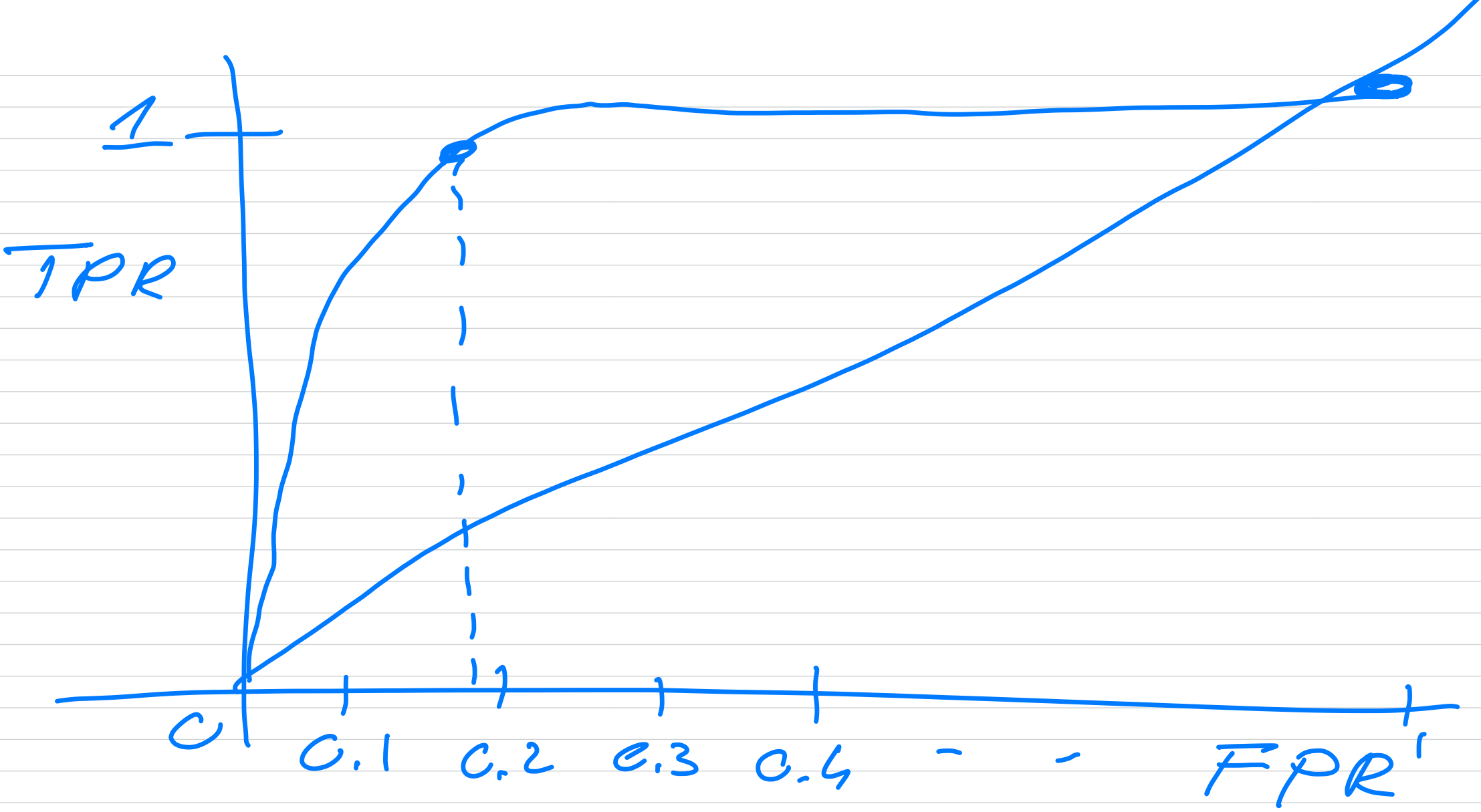
$$TNR = \frac{TN}{TN + FP} = 1 - FPR$$

ROC-curve FPR vs TPR

(0,0) Flags nobody

(1,1) Flags everybody

A perfect ranking goes through (0,1), a random is the diagonal


1
TPR
0
0.1
0.2
0.3
0.4
FPR